

EE3621: Digital Signal Processing

Student Lecture Notes — Lecture 26 Equalization, Noise Cancellation & Adaptive FIR Filters

III B.Tech EEE — Core Exam Preparation Guide

1 Channel Equalization & The Zero-Forcing Equalizer

1.1 Inter-Symbol Interference (ISI)

When digital symbols $s[n]$ are transmitted across a dispersive communication channel $h_c[n]$, multipath delays spread out adjacent symbols, creating **Inter-Symbol Interference (ISI)**:

$$x[n] = s[n] \otimes h_c[n] + v[n] = h_c[0]s[n] + \sum_{k \neq 0} h_c[k]s[n-k] + v[n] \quad (1)$$

1.2 The Zero-Forcing Equalizer (ZFE) Criterion & Matrix Formulation

An equalizing filter $h_{eq}[n]$ inverts channel distortion so that the combined response is a pure delay of Δ samples:

$$H_{eq}(z) \cdot H_c(z) = z^{-\Delta} \iff \mathbf{q}[n] = \mathbf{h}_c[n] \otimes \mathbf{h}_{eq}[n] = \begin{cases} 1, & n = \Delta \\ 0, & n \neq \Delta \end{cases} \quad (2)$$

For an N -tap FIR equalizer $\mathbf{w} = [w_0, w_1, \dots, w_{N-1}]^T$ and a channel $h_c[n]$ of length L , the convolution $q[n] = \sum_{k=0}^{N-1} w_k h_c[n-k]$ yields the linear matrix equation:

$$\mathbf{H}_c \mathbf{w} = \mathbf{u}_\Delta \implies \begin{bmatrix} h_c[0] & 0 & \dots & 0 \\ h_c[1] & h_c[0] & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ h_c[L-1] & h_c[L-2] & \dots & h_c[L-N] \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_{N-1} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \quad (3)$$

where $\mathbf{u}_\Delta$ has a single 1 at row index Δ . For an FIR channel, an exact IIR inverse is required; a finite FIR equalizer forces the first N samples of $q[n]$ to match $\mathbf{u}_\Delta$, leaving a small residual ISI tail.

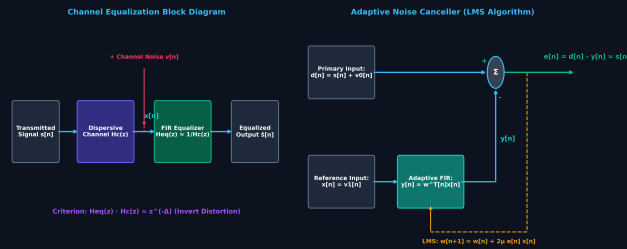


Figure 1: Left: Channel Equalizer inverting dispersive channel $H_c(z)$. Right: Dual-input Adaptive Noise Canceller driven by the LMS algorithm.

2 Adaptive Noise Cancellation (ANC)

2.1 Dual-Input Architecture

Unlike conventional static filtering, Adaptive Noise Cancellation uses two separate inputs:

- **Primary Input** $d[n]$: Desired signal $s[n]$ contaminated by noise $v_0[n]$:

$$d[n] = s[n] + v_0[n] \quad (4)$$

- **Reference Input** $x[n]$: Noise $v_1[n]$ that is correlated with $v_0[n]$, but **completely uncorrelated with the signal** $s[n]$ ($E[s[n] \cdot v_1[n - k]] = 0$).

2.2 Mathematical Proof of Signal Extraction

The system error output is the difference between primary input and filter estimate $y[n]$:

$$e[n] = d[n] - y[n] = s[n] + (v_0[n] - y[n]) \quad (5)$$

Squaring and taking the statistical expectation:

$$E[e^2[n]] = E[s^2[n]] + E[(v_0[n] - y[n])^2] + 2E[s[n](v_0[n] - y[n])] \quad (6)$$

Since $s[n]$ is uncorrelated with both $v_0[n]$ and $y[n]$, the cross-term is zero:

$$\mathbf{E}[e^2[\mathbf{n}]] = \mathbf{E}[s^2[\mathbf{n}]] + \mathbf{E}[(\mathbf{v}_0[\mathbf{n}] - \mathbf{y}[\mathbf{n}])^2] \quad (7)$$

The Core Principle: Signal power $E[s^2[n]]$ is unaffected as filter weights adapt. Therefore, minimizing total output power $E[e^2[n]]$ forces:

$$y[n] \longrightarrow v_0[n] \implies \mathbf{e}[\mathbf{n}] \longrightarrow \mathbf{s}[\mathbf{n}] \quad (8)$$

The system error output $e[n]$ becomes the ****clean, noise-free signal****!

3 The Adaptive FIR Filter & The LMS Algorithm

3.1 Derivation of the LMS Weight Update Equation

Let the Mean-Squared Error (MSE) cost function be $J(\mathbf{w}) = E[e^2[n]]$. The filter output is $y[n] = \mathbf{w}^T[n]\mathbf{x}[n]$, giving error $e[n] = d[n] - \mathbf{w}^T[n]\mathbf{x}[n]$. Computing the gradient of $J(\mathbf{w})$ with respect to the weight vector $\mathbf{w}$:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \nabla_{\mathbf{w}} E[e^2[n]] = 2E[e[n]\nabla_{\mathbf{w}} e[n]] \quad (9)$$

Evaluating $\nabla_{\mathbf{w}} e[n] = \nabla_{\mathbf{w}} (d[n] - \mathbf{w}^T \mathbf{x}[n]) = -\mathbf{x}[n]$:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = -2E[e[n]\mathbf{x}[n]] \quad (10)$$

The classic method of ****Steepest Descent**** updates weights in the negative gradient direction:

$$\mathbf{w}[n+1] = \mathbf{w}[n] - \frac{1}{2}\mu \nabla_{\mathbf{w}} J = \mathbf{w}[n] + \mu E[e[n]\mathbf{x}[n]] \quad (11)$$

Widrow–Hoff Stochastic Approximation: To eliminate computing statistical expectations in real time, the Least Mean Squares (LMS) algorithm replaces the ensemble expectation $E[e[n]\mathbf{x}[n]]$ with the ****instantaneous sample estimate**** $e[n]\mathbf{x}[n]$:

$$\mathbf{w}[\mathbf{n}+1] = \mathbf{w}[\mathbf{n}] + 2\mu \mathbf{e}[\mathbf{n}] \mathbf{x}[\mathbf{n}] \quad (12)$$

3.2 The 3 Core Operational Equations of LMS

For an M -tap adaptive transversal FIR filter:

1. **Filter Output:** $y[n] = \mathbf{w}^T[n]\mathbf{x}[n] = \sum_{k=0}^{M-1} w_k[n]x[n-k]$
2. **Error Estimation:** $e[n] = d[n] - y[n]$
3. **Weight Vector Update:** $\mathbf{w}[n+1] = \mathbf{w}[n] + 2\mu e[n] \mathbf{x}[n]$

3.3 Step-Size Parameter μ Stability Criterion

To guarantee convergence without divergent numerical instability:

$$0 < \mu < \frac{1}{M \cdot P_x} \quad (13)$$

where M is the number of filter taps and $P_x = E[x^2[n]]$ is the input signal power.

4 Standard Solved Exam Examples

Solved Exam Example 1: 2-Tap Adaptive LMS Filter Iterations

Problem 1: A 2-tap adaptive FIR filter has initial weight vector $\mathbf{w}[0] = [0.0, 0.0]^T$ and adaptation step-size $\mu = 0.1$. Given the input sequences:

- At $n = 0$: Reference vector $\mathbf{x}[0] = [1.0, 0.5]^T$, Primary target $d[0] = 2.0$.
- At $n = 1$: Reference vector $\mathbf{x}[1] = [0.5, 1.0]^T$, Primary target $d[1] = 1.5$.

Compute the filter output $y[n]$, error $e[n]$, and updated weight vector $\mathbf{w}[n+1]$ for iterations $n = 0$ and $n = 1$. Verify the convergence stability bound.

Solution:

Iteration 1 ($n = 0$):

1. **Filter Output $y[0]$:**

$$y[0] = \mathbf{w}^T[0]\mathbf{x}[0] = w_0[0]x[0] + w_1[0]x[-1] = (0.0)(1.0) + (0.0)(0.5) = \mathbf{0.0} \quad (14)$$

2. **Estimation Error $e[0]$ and Instantaneous Square Error:**

$$e[0] = d[0] - y[0] = 2.0 - 0.0 = \mathbf{2.0}, \quad e^2[0] = (2.0)^2 = \mathbf{4.00} \quad (15)$$

3. **Weight Vector Update $\mathbf{w}[1]$:** With $2\mu = 2(0.1) = 0.2$, the scalar update factor is $2\mu e[0] = 0.2 \times 2.0 = \mathbf{0.40}$:

$$\mathbf{w}[1] = \mathbf{w}[0] + 2\mu e[0]\mathbf{x}[0] = \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix} + 0.40 \begin{bmatrix} 1.0 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 0.40 \\ 0.20 \end{bmatrix} \quad (16)$$

Iteration 2 ($n = 1$):

1. **Filter Output $y[1]$:**

$$y[1] = \mathbf{w}^T[1]\mathbf{x}[1] = w_0[1]x[1] + w_1[1]x[0] = (0.40)(0.5) + (0.20)(1.0) = 0.20 + 0.20 = \mathbf{0.40} \quad (17)$$

2. **Estimation Error $e[1]$ and Instantaneous Square Error:**

$$e[1] = d[1] - y[1] = 1.5 - 0.40 = \mathbf{1.10}, \quad e^2[1] = (1.10)^2 = \mathbf{1.21} \quad (18)$$

Convergence Check: Squared error dropped from $e^2[0] = 4.00$ to $e^2[1] = 1.21$, achieving a **69.75%** reduction in MSE in just a single step!

3. **Weight Vector Update $\mathbf{w}[2]$:** Scalar update factor: $2\mu e[1] = 0.2 \times 1.10 = \mathbf{0.22}$:

$$\mathbf{w}[2] = \mathbf{w}[1] + 2\mu e[1]\mathbf{x}[1] = \begin{bmatrix} 0.40 \\ 0.20 \end{bmatrix} + 0.22 \begin{bmatrix} 0.5 \\ 1.0 \end{bmatrix} = \begin{bmatrix} 0.51 \\ 0.42 \end{bmatrix} \quad (19)$$

4. **Step-Size Stability Check:** Average input power $P_x \approx \frac{1}{2}(1.0^2 + 0.5^2) = \frac{1.25}{2} = 0.625$. Upper stability bound:

$$\mu_{\max} = \frac{1}{M \cdot P_x} = \frac{1}{2 \times 0.625} = \mathbf{0.80} \quad (20)$$

Since selected $\mu = 0.1 \ll 0.80$, the algorithm is unconditionally stable.

Solved Exam Example 2: Zero-Forcing Equalizer (ZFE) Design

Problem 2: A multipath communication channel has the discrete impulse response $h_c[n] = \{1.0, 0.4\}$. Design a 3-tap FIR Zero-Forcing Equalizer $\mathbf{w} = [w_0, w_1, w_2]^T$ to eliminate ISI with zero delay ($\Delta = 0$). Determine the combined equalized impulse response $q[n]$ and calculate the percentage ISI reduction.

Solution:

Step 1: Set Up Convolution Equations

The combined response is $q[n] = h_c[n] \otimes w[n] = \sum_{k=0}^2 w_k h_c[n-k]$. For zero delay ($\Delta = 0$), the zero-forcing condition enforces $q[0] = 1$ and $q[1] = q[2] = 0$:

$$\begin{aligned} n = 0 : \quad q[0] &= h_c[0]w_0 = 1.0 \implies 1.0w_0 = 1.0 \implies \mathbf{w_0 = 1.00} \\ n = 1 : \quad q[1] &= h_c[1]w_0 + h_c[0]w_1 = 0 \implies 0.4(1.0) + 1.0w_1 = 0 \implies \mathbf{w_1 = -0.40} \\ n = 2 : \quad q[2] &= h_c[1]w_1 + h_c[0]w_2 = 0 \implies 0.4(-0.40) + 1.0w_2 = 0 \implies \mathbf{w_2 = +0.16} \end{aligned} \quad (21)$$

The designed equalizer tap weight vector is:

$$\mathbf{w} = [\mathbf{1.00}, \mathbf{-0.40}, \mathbf{0.16}]^T \quad (22)$$

Step 2: Evaluate the Residual Equalizer Tail Because the equalizer is finite-length ($N = 3$), a non-zero tail remains at $n = 3$:

$$q[3] = h_c[1]w_2 = 0.4 \times 0.16 = \mathbf{0.064} \quad (23)$$

The complete equalized channel response is:

$$\mathbf{q[n] = \{1.000, 0.000, 0.000, 0.064\}} \quad (24)$$

Step 3: Quantitative ISI Reduction

- Unequalized channel peak ISI: $|h_c[1]| = \mathbf{0.400}$.
- Equalized channel residual ISI: $|q[3]| = \mathbf{0.064}$.
- Percentage ISI reduction:

$$\text{ISI Reduction} = \frac{0.400 - 0.064}{0.400} \times 100\% = \frac{0.336}{0.400} \times 100\% = \mathbf{84.0\%} \quad (25)$$

5 Essential Exam Checklist & Traps

- ☐ **LMS Order of Operations:** Always compute in strict order: $y[n] \rightarrow e[n] \rightarrow \mathbf{w}[n+1]$. Never update weights before computing the output!
- ☐ **The Factor of 2 in LMS:** Remember the update is $\mathbf{w}[n+1] = \mathbf{w}[n] + 2\mu e[n]\mathbf{x}[n]$.
- ☐ **Reference Input Isolation:** In ANC, the reference input $x[n]$ must contain ****only noise****; it must never contain the desired signal $s[n]$.
- ☐ **Step-Size Bounds:** Selecting $\mu > 1/(MP_x)$ leads to immediate mathematical divergence ($w_k \rightarrow \infty$).